Table 1. **Coding-agent baseline performance on KramaBench with oracle file selection (N=94 tasks).** To establish an upper-bound reference for code generation and execution independently of file retrieval, we evaluated our coding agent on KramaBench—a public general-purpose data-science benchmark of diverse tabular reasoning tasks—under an oracle condition in which all task-relevant files are pre-attached to the model, bypassing the dataset-discovery stage entirely. *Score%* is the composite KramaBench score. *Exact (Reports)* is the fraction of all 94 tasks for which the agent produced the exact correct answer. *Exact (Executed)* is the exact-match rate conditioned on tasks that reached successful execution, isolating answer-quality from execution failures. *Executed%* is the fraction of tasks for which the generated script ran successfully in the sandbox and produced any answer, regardless of correctness; values of 61.7%–78.7% across model backends demonstrate that code generation and execution are performant when provided with correct inputs. Results across four model backends (gpt-5.3-codex, gpt-5.4-03-2026, gpt-5.4-mini, gpt-5.4-nano) confirm that execution success is consistently high under oracle conditions, and that the bottleneck in real-world deployment lies in file retrieval rather than code generation per se (cf. Table 2).

Model	Score %	Exact Count	Exact% (Rep.)	Exact% (Exec.)	Exec. Count	Exec. %
gpt-5.3-codex	42.18	30/94	31.9%	40.5%	74/94	78.7%
gpt-5.4-03-2026	40.79	27/94	28.7%	39.1%	69/94	73.4%
gpt-5.4-mini	32.04	24/94	25.5%	40.0%	60/94	63.8%
gpt-5.4-nano	29.90	19/94	20.2%	32.8%	58/94	61.7%

Table 2. **Coding-agent end-to-end performance on KramaBench with autonomous dataset discovery (N=106 tasks, discovery step budget = 6).** Unlike the oracle baseline in Table 1, the agent here must autonomously search a raw dataset root—with no prior knowledge of which files are relevant—finalize a file selection, and then generate and execute code, all within a fixed step budget. This end-to-end condition tests file-retrieval and code-generation capabilities jointly and represents the realistic deployment setting. Column definitions are identical to Table 1. Execution rates drop to 29.2%–39.6% compared with 61.7%–78.7% under oracle conditions, a reduction of 24–44 percentage points that is attributable to tasks that fail at the discovery stage and never reach code execution. Notably, gpt-5.4-mini and gpt-5.4-nano incur higher token costs than stronger models despite similar or lower execution rates, because these models frequently exhaust the full discovery budget without converging to a valid file selection, wasting inference compute at the retrieval stage. Together with Table 1, these results isolate dataset discovery as the primary performance bottleneck: code generation and execution are competitive when the correct files are provided, but autonomous file retrieval remains the limiting factor for real-world deployment.

Model	Score%	Exact	Exact% (Reports)	Exact% (Executed)	Executed	Executed%
gpt-5.3-codex	18.28	13/106	12.3%	32.5%	40/106	37.7%
gpt-5.4-03-2026	15.76	10/106	9.4%	32.3%	31/106	29.2%
gpt-5.4-mini	12.89	10/106	9.4%	23.8%	42/106	39.6%
gpt-5.4-nano	12.96	10/106	9.4%	25.0%	40/106	37.7%

Table 3. **Three-stage task attrition under autonomous dataset discovery on KramaBench (N=106 tasks per model).** To identify *where* tasks fail in the end-to-end pipeline, we decompose each model's 106 tasks into three sequential stages: (1) *file selection*: discovery successfully finalizes at least one relevant file within the step budget; (2) *execution*: the generated script runs in the sandbox and produces an output; (3) *exact match*: the output matches the gold answer. Tasks that fail at any stage do not proceed to subsequent stages, so attrition is cumulative. The *Dominant failure mode* column identifies the pipeline stage responsible for the greatest absolute task loss per model. gpt-5.3-codex and gpt-5.4-03-2026 lose the majority of tasks at the file-selection stage (60.4% and 68.9% of tasks never produce a valid file set, respectively), confirming that strong reasoning models are bottlenecked by retrieval, not by code generation. In contrast, gpt-5.4-mini selects files for 63.2% of tasks but then fails disproportionately at the execution stage, indicating that aggressive file selection produces noisier inputs that impede successful script execution. Across all models, exact-match rates remain low (9.4%–12.3%), underscoring that improving dataset discovery is the primary lever for advancing overall end-to-end performance.

Model	Files selected	Selected%	Executed	Executed%	Exact	Exact%	Dominant failure
gpt-5.3-codex	42/106	39.6%	40/106	37.7%	13/106	12.3%	No files selected
gpt-5.4-03-2026	33/106	31.1%	31/106	29.2%	10/106	9.4%	No files selected
gpt-5.4-mini	67/106	63.2%	42/106	39.6%	10/106	9.4%	Script exec. failure
gpt-5.4-nano	52/106	49.1%	40/106	37.7%	10/106	9.4%	No files selected

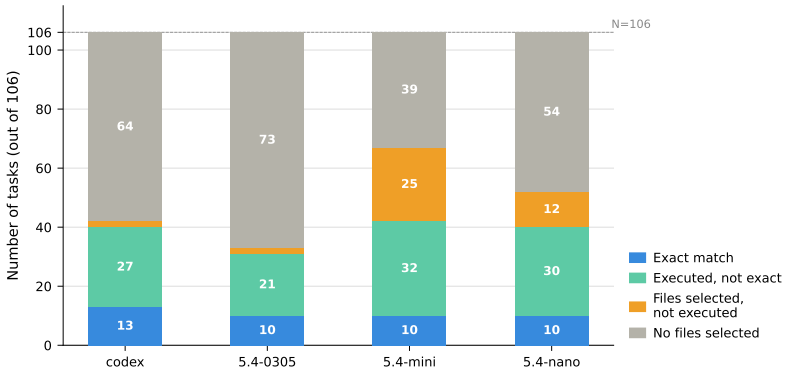

Figure 1. **Task attrition across the three pipeline stages of the autonomous dataset-discovery workflow on KramaBench (N=106 tasks per model).** The coding agent must autonomously identify relevant files from a raw dataset root, generate a solution script, and execute it in a sandboxed environment; tasks that fail at any stage do not proceed further. Each stacked bar shows how 106 tasks are distributed across four mutually exclusive outcomes for a given model backend: **exact match**: script executed and produced the correct answer; **executed, not exact**: script ran but produced an incorrect answer; **files selected, not executed**: discovery succeeded but the script failed or was not attempted; **no files selected**: discovery exhausted its step budget without finalizing any file set. Numbers inside bars are task counts. The dominant gray segment for gpt-5.3-codex and gpt-5.4 shows that these stronger models fail primarily at the discovery stage (60.4% and 68.9% of tasks, respectively), not at code generation. gpt-5.4-mini reaches execution for the largest fraction of tasks (39.6%) by being more aggressive at file selection (63.2% file selection rate), but sustains proportionally higher execution losses, revealing a trade-off between retrieval recall and execution precision. Across all models, the largest single failure category is *no files selected*, establishing dataset discovery as the primary bottleneck for end-to-end performance improvement (cf. Tables 2–3).

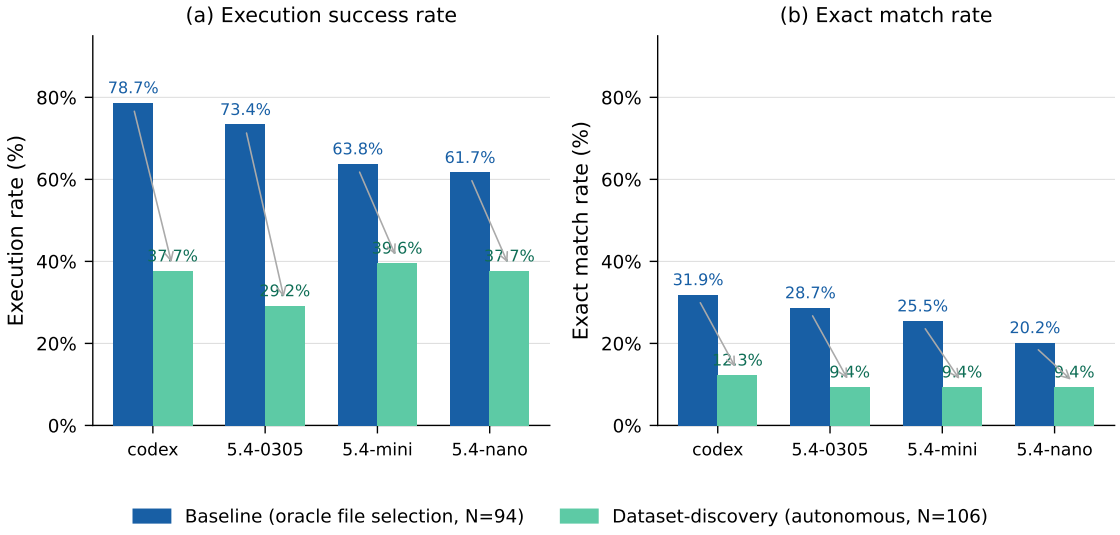

Figure 2. **Impact of autonomous dataset discovery on execution success and exact-match rates across four model backends on KramaBench.** We compare two conditions to isolate the contribution of dataset discovery to end-to-end performance. **Dark blue** bars (*oracle baseline*, N=94 tasks): task-relevant files are pre-attached to the model, eliminating the discovery stage and providing an upper-bound reference for code generation and execution in isolation. **Teal** bars (*autonomous discovery*, N=106 tasks): the agent must search the raw dataset root and finalize a file selection before coding, representing the realistic deployment setting. **(a)** Execution success rate: fraction of tasks for which the generated script ran and produced any answer. **(b)** Exact-match rate: fraction of tasks whose output matched the gold answer. Arrows indicate the per-model drop from baseline to discovery. Execution rates fall by 24–44 percentage points under autonomous discovery, while exact-match rates fall by 10–22 percentage points. Because code generation and execution are competitive under oracle conditions (61.7%–78.7% execution rate), the gap between the two conditions directly quantifies the performance cost of autonomous file retrieval. Improving dataset discovery is therefore the primary lever for closing the gap between controlled and real-world coding-agent performance.

Table 4. **Comprehensive comparison of hypothesis-generation pipelines across automated LLM evaluation and blinded expert oncologist assessment ($n = 10$ runs per pipeline).** *Panel A: Automated novelty evaluation.* Each hypothesis was evaluated by an LLM-based multi-agent ensemble that independently queried three scientific databases (Semantic Scholar, PubMed, arXiv). *Mean novelty score* reports the average rating assigned by all ensemble agents on a 1–10 scale. *Consensus score* measures inter-agent agreement within the ensemble on the same 1–10 scale; higher values indicate that agents converge on the same novelty assessment. *Evaluator confidence* is the ensemble's self-reported certainty (0–100 %) in the aggregated verdict; lower values suggest that the hypothesis is unusual relative to retrievable prior work, which is consistent with higher genuine novelty. *References retrieved*$^{\downarrow}$ counts the distinct literature sources the evaluation ensemble retrieved when searching for prior work similar to each hypothesis. A lower count indicates that the hypothesis has fewer close matches in the indexed literature, serving as a proxy for novelty: hypotheses that are harder to corroborate through retrieval are more likely to explore territory not yet covered by published work. *Panel B: Blinded expert oncologist evaluation.* A board-certified oncologist with *X* years of clinical and translational research experience independently scored each hypothesis on six equally weighted criteria ($w = 1/6$) using a three-point ordinal scale (1 = Poor, 3 = Moderate, 5 = Excellent), with no knowledge of which pipeline generated each hypothesis. The final score is the unweighted mean across all six criteria, $S = \frac{1}{6}\sum_{i=1}^{6} s_i$. All values are mean [95 % CI] across $n = 10$ independent runs (Student's t-distribution, $\mathrm{df} = 9$). Statistical comparisons between SAGE and each baseline were performed using two-sided Mann–Whitney U tests (non-parametric; appropriate for ordinal and small-sample data). * denotes $p < 0.05$ vs. both baselines simultaneously; † denotes $p < 0.05$ vs. ChatGPT only (difference vs. Agent Flow did not reach significance at $n = 10$). **Bold** entries indicate the best-performing pipeline per metric. $^{\downarrow}$Lower values are preferable for this metric. ‡All 10 SAGE hypotheses received a Robustness score of 3 (zero variance; 95 % CI not applicable).

Metric	ChatGPT	Agent Flow	SAGE (ours)
Panel A — Automated LLM novelty evaluation			
Novelty score (1–10)	7.04 [6.57, 7.51]	7.10 [6.73, 7.47]	**7.27 [6.53, 8.01]**
Consensus score (1–10)	7.10 [6.47, 7.73]	7.10 [6.69, 7.51]	**7.30 [6.54, 8.06]**
Evaluator confidence (%)	**84.20 [77.04, 91.36]**	83.00 [76.72, 89.28]	77.60 [70.25, 84.95]
Refs. retrieved$^{\downarrow}$	6.00 [5.25, 6.75]	5.20 [4.20, 6.20]	**2.40 [1.17, 3.63]***
Panel B — Blinded expert oncologist evaluation (scale 1–5)			
Novelty	1.20 [0.75, 1.65]	1.80 [1.06, 2.54]	**4.00 [3.25, 4.75]***
Plausibility	**3.00 [2.33, 3.67]**	2.40 [1.71, 3.09]	**3.20 [2.75, 3.65]**
Feasibility	2.60 [1.70, 3.50]	3.00 [2.33, 3.67]	**3.40 [2.80, 4.00]**
Implementability	1.80 [1.06, 2.54]	2.00 [1.25, 2.75]	**3.00 [2.05, 3.95]**†
Clinical Impact	1.40 [0.80, 2.00]	2.40 [1.43, 3.37]	**3.60 [2.63, 4.57]**†
Robustness	2.80 [2.35, 3.25]	2.60 [2.00, 3.20]	**3.00**‡
Final score (1–5)	2.13 [1.83, 2.43]	2.37 [1.91, 2.82]	**3.37 [2.95, 3.78]***

Revised Figure and Table Captions

Figure 1. **Overview of the SAGE framework for end-to-end computational pathology biomarker discovery.** SAGE operates through three tightly coupled stages. *(1) Knowledge graph construction and reasoning* (bottom left): scientific literature spanning bladder cancer and computational pathology is ingested, chunked, and processed by an LLM to extract biomedical triples of the form (head entity, relation, tail entity). Extracted triples are assembled into local per-document graphs that are subsequently fused into a single global knowledge graph encoding biological mechanisms, clinical endpoints, and computational methods. *(2) Hypothesis generation and refinement* (top): a fixed-order sequence of specialized agents operates as follows. A *Path Generation Agent* traverses the knowledge graph to identify multi-hop paths linking biomedical entities. An *Ontologist* grounds each path in curated biomedical ontologies, mapping symbolic relations to biologically interpretable processes. A *Scientist* agent formulates concise, dataset-aware hypotheses from the ontology-enriched paths. A human-in-the-loop checkpoint then allows domain experts to approve, reject, or refine hypotheses before downstream processing. A *Hypothesis Expansion Agent* enriches the approved hypothesis with biological rationale and clinical context. A *Novelty Critic* evaluates originality by retrieving and contrasting existing literature. A *Feasibility Agent* confirms that the hypothesis is testable with available data, tools, and computational resources. *(3) Validation and summarization* (bottom right): a *Coding Tasks Agent* translates the hypothesis into structured coding experiments. A *Coding Agent* prepares data, generates and iteratively debugs solution code, and executes it in a sandboxed environment via a Tool Orchestration Subsystem (detailed in Figure 5) that provides unified access to histopathology and clinical oncology tools. A *Coding Results Interpreter* analyzes the resulting plots, metrics, and tables to assess hypothesis validity. A *Summary Agent* consolidates the hypothesis, novelty score, feasibility score, and supporting rationale into a clinically interpretable report. Together, these stages form a closed discovery loop from literature-grounded hypothesis generation through empirical validation.

Figure 2. **Visualization of the expert-curated biomedical knowledge graph used for hypothesis generation in SAGE.** The graph was constructed by extracting and fusing biomedical triples from 1,650 scientific sources comprising 1,000 bladder cancer papers, 650 computational pathology papers, and three reference textbooks, yielding a final graph of 41,053 nodes and 56,338 edges. Each node represents a distinct biomedical entity such as a gene, disease, pathway, phenotype, tissue region, clinical endpoint, staining method, or computational algorithm; node colors encode entity type according to the legend. Each edge encodes a semantic or mechanistic relationship mined from the literature and normalized via ontology-aware rule-based mapping, with edge weights reflecting extraction confidence. The main panel shows the full global graph, illustrating the heterogeneous and densely connected nature of the biomedical knowledge space captured by SAGE. The zoomed-in subgraph (bottom right) highlights a bladder cancer-centric region, revealing dense multi-scale connectivity among genes, pathways, tissue phenotypes, and clinical outcomes that enables context-aware, multi-hop reasoning during hypothesis generation.

Figure 3.**Pathology-informed overall survival analysis for the FABP5–TLS joint stratification hypothesis in TCGA-BLCA, generated and validated end-to-end by SAGE.** SAGE autonomously identified a multi-modal prognostic interaction between FABP5 (E-FABP) RNA expression, a fatty acid binding protein with roles in lipid metabolism and immune modulation, and the abundance of tertiary lymphoid structure (TLS)-like lymphoid aggregates quantified from whole-slide H&E images. Patients in TCGA-BLCA were jointly stratified into four groups by FABP5 expression level (high vs. low, dichotomized at the median) and TLS-like aggregate density (abundant vs. scarce, derived from automated whole-slide image analysis): high FABP5/scarce TLS (red), high FABP5/abundant TLS (orange), low FABP5/scarce TLS (blue), and low FABP5/abundant TLS (green). Kaplan–Meier survival curves are shown with the number of patients at risk reported at each time point. Log-rank testing across all four groups yielded p=0.107p = 0.107 p=0.107; the targeted pairwise comparison between the high-FABP5/scarce-TLS and low-FABP5/abundant-TLS groups demonstrated statistically significant survival separation (p=0.0129p = 0.0129 p=0.0129). Cox proportional hazards modeling confirmed elevated mortality risk in the high-FABP5/scarce-TLS group relative to the low-FABP5/abundant-TLS reference (HR = 2.34, 95 % CI [1.25, 4.40]). This result illustrates SAGE's capacity to discover biologically interpretable, multi-modal prognostic biomarkers that are statistically supported and immediately actionable using publicly available cohort data.

Table 1.**Computational efficiency comparison of the graph-based specific-memory pipeline (GP) and the chat-based shared-memory pipeline (CP) on TCGA-BLCA hypothesis generation runs.** In the GP architecture, each agent receives only the task-relevant output of its immediate predecessor, minimizing redundant context and token consumption. In the CP architecture, all agents have access to the full conversation history, which increases context length but introduces redundant information. Both pipelines were evaluated across multiple independent runs on TCGA-BLCA. *Prompt tokens* and *completion tokens* report the mean number of input and output tokens consumed per run (mean $\pm$ SD). *Time (sec.)* reports mean wall-clock runtime per run (mean $\pm$ SD). *Novelty* reports the hypothesis novelty score assigned by the multi-critic evaluation ensem-

ble on a 1–10 scale. *Relative cost* normalizes inference cost to the GP baseline ($1\times$). The GP pipeline reduced prompt token usage by 57.5% and completion token usage by 14.9%, corresponding to a 25–75% reduction in inference cost, while achieving comparable runtime and only a marginal reduction in novelty score (7/10 vs. 8/10). These results demonstrate that targeted context allocation in GP preserves hypothesis quality while enabling substantially more cost-efficient and predictable execution at scale.

Table 2.**Automated novelty evaluation of hypothesis-generation pipelines on the TLS–gene bladder cancer task ($n = 10$ runs per pipeline; values reported as mean $\pm$ SD).** Three pipelines were compared under identical prompting conditions using the task: *"Identify a novel research hypothesis linking tertiary lymphoid structures (TLS) and genes in bladder cancer."* *ChatGPT* denotes a single-response LLM baseline (ChatGPT 5.1) that generates one hypothesis per run without iterative refinement or literature retrieval. *Agent Flow* (**?**) is a multi-agent iterative pipeline with explicit web browsing at each step. *SAGE (ours)* generates hypotheses through structured traversal of a domain-specific biomedical knowledge graph without exhaustive external retrieval during generation. All 30 generated hypotheses were evaluated using the same LLM-based multi-agent novelty scoring ensemble, which independently queried Semantic Scholar, PubMed, and arXiv. *Novelty (mean rev.)* is the mean novelty score assigned across all ensemble agents on a 1–10 scale. *Consensus* is the degree of inter-agent agreement within the ensemble on the same scale. *Refs. retrieved* is the number of distinct literature sources collectively retrieved and cited during evaluation; a lower count indicates that the hypothesis is less directly recoverable from indexed prior work, reflecting greater specificity and distance from well-established associations. SAGE achieves the highest novelty and consensus scores while retrieving substantially fewer references, demonstrating that structured knowledge-graph reasoning produces hypotheses that are comparably novel to retrieval-heavy pipelines but more deeply grounded in underexplored biological territory.

Table 3. **Effect of adversarial multi-critic deliberation on novelty assessment accuracy and score separability across a five-tier novelty spectrum (150 scientific proposals, 30 per tier).** Four evaluation configurations are compared to isolate the incremental contribution of critic diversity and disagreement-triggered deliberation. *Single-Critic*: one agent scores each proposal independently, providing a lower-bound reference. *Two-Critic (Verifier + Prover)*: a Prover (optimistic) and a Verifier (conservative) score independently without interaction; scores are averaged. *Multi-Critic (No Debate)*: all three critics (Prover, Verifier, and Judge) score independently and scores are averaged, with no iterative argumentation. *Multi-Critic (Full)*: identical to the no-debate condition except that when initial inter-critic disagreement exceeds a threshold (SD > 1.0), critics enter an iterative Bayesian debate in which the Prover presents novelty claims, the Verifier retrieves contradicting literature, and the Judge synthesizes a final verdict; debate continues until convergence (SD < 0.5) or a maximum of three rounds. *Accuracy* is the fraction of proposals correctly classified into their ground-truth novelty tier. *A–E Gap* is the mean score difference between breakthrough proposals (Tier A) and specious proposals (Tier E), measuring the system's ability to discriminate across the full novelty spectrum. All proposals were evaluated using time-travel backtesting against literature available only prior to publication or retraction, preventing temporal leakage. Enabling full deliberation yields the largest marginal gain in both accuracy (+10.0 percentage points over no-debate) and A–E separability (+1.76 points), demonstrating that disagreement-triggered debate improves calibration beyond what critic diversity or score averaging alone can achieve.

Figure 4. **Overview of the three-phase AI-enabled literature processing and knowledge graph construction workflow used to build the SAGE biomedical knowledge base.** *Phase 1 – Multi-source keyword expansion:* starting from a seed set of domain keywords provided by subject matter experts (e.g., Bladder Cancer", Breast Cancer"), an iterative AI expansion step using OpenAI, MESH, and Semantic Scholar APIs generates an enlarged keyword set. For bladder cancer, 308 base keywords were expanded to 1,276 terms; after intelligent GPT-based filtering for relevance and specificity, 748 refined BLCA keywords were retained for crawling. *Phase 2 – Large-scale literature crawling:* the refined keyword set is used to query high-impact journals and biomedical repositories using 30–50 multithreaded parallel connections, yielding 252,614 unique bladder cancer DOIs (137,000 papers retrieved) and 674,150 unique breast cancer DOIs (233,000 papers retrieved plus NCCN guidelines). *Phase 3 – Automated article processing and storage:* retrieved papers are processed via two paths. Path A (bladder cancer) prioritizes direct text extraction from PDFs using a three-tier fallback strategy; if extraction fails, the compressed PDF is stored. Path B (breast cancer) applies optimized step-wise PDF compression to reduce storage requirements. NCCN clinical guideline graphics are converted to structured text using GPT-based graphics-to-text conversion. Processed datasets are stored in a scalable cloud workflow and are then ready for triple extraction and knowledge graph construction.

Table 4. **Quality assessment of the SAGE biomedical knowledge graph before and after ontology refinement, evaluated on a stratified random sample of 200 extracted triples.** Triple quality was assessed along three dimensions. *Factual grounding* measures whether the extracted triple is supported by the cited evidence span in the source document. *Relation accuracy* measures whether the extracted

relation phrase correctly captures the direction and semantics of the stated relationship between head and tail entities. *Entity type accuracy* measures whether the ontology type assigned to each entity (e.g., Gene, Disease, Pathway, ClinicalEndpoint) is correct; strict accuracy requires an exact type match, while lenient accuracy accepts semantically compatible types. *Hallucination rate* is the fraction of triples containing fabricated or unsupported information not present in the source text. The initial pipeline achieved high factual grounding (99.0%) and perfect relation accuracy (100.0%), but lower entity type accuracy (64.6% strict) due to systematic ontology ambiguities in complex noun phrases and clinical endpoint disambiguation. Following ontology refinement and post-processing validation, strict entity type accuracy improved to 82.5% and the hallucination rate remained below 1.0%, confirming the reliability of the knowledge graph as a foundation for biological reasoning and hypothesis generation.

Table 5. **Complete agent workflow for the SAGE hypothesis generation pipeline, specifying the role, context source, assigned model, and model-selection rationale for each agent operating under the graph-based specific-memory architecture (GP).** In the GP architecture, each agent receives only the output of its immediate predecessor rather than the full conversation history, minimizing redundant token usage and context dilution. The *Path Generation Agent* initiates the pipeline by querying and traversing the knowledge graph; it is assigned GPT-5 Nano, the lowest-cost model sufficient for structured graph traversal with minimal free-form reasoning. The *Ontologist* maps symbolic relations to biologically interpretable processes and is assigned GPT-5 Mini, which provides sufficient domain knowledge without the cost of the full reasoning model. The *Scientist* formulates a concise, empirically testable hypothesis and is assigned the highest-capability GPT-5 model with high-reasoning mode, as this is the most critical reasoning step in the pipeline. The *Hypothesis Expansion Agent* enriches the hypothesis with biological rationale using GPT-5 Mini, where clarity is prioritized over maximal reasoning depth. The *Novelty Critic Agent* searches online resources for similar publications and assigns a novelty score on a 1–10 scale using GPT-5 Mini. The *Coding Instructions Agent* translates the hypothesis into detailed, dataset-aware coding specifications using GPT-5 with high reasoning, as precision here is critical for effective downstream validation. The *Coding Results Interpreter* analyzes plots, statistical outputs, and tables from code execution using GPT-5 with high reasoning and visual tool access to support multimodal interpretation. The *Summary Agent* synthesizes the final hypothesis, novelty score, and feasibility score into a concise report using GPT-5 Mini. The Coding Agent sub-system (Task Agent, Data Agent, Validation Agent) is described separately in Section 3.3 of the main text.

Table 6. **Feasibility Agent evaluation protocol and weighted scoring criteria used to assess whether a generated hypothesis is practically testable prior to computational validation.** The Feasibility Agent applies a four-step verification protocol to each candidate hypothesis, producing a weighted feasibility score and a categorical verdict (feasible, conditionally feasible, or infeasible). *Data Reality Check* (weight 35%) verifies that suitable datasets exist with adequate sample sizes and that all required variables, labels, and metadata are available for the proposed analysis. *Tech-Stack Alignment* (weight 25%) assesses whether mature, actively maintained analysis tools and reference implementations exist for the required computational methods. *Logical Stress Test* (weight 25%) identifies gaps in causal reasoning, potential confounders, and whether the hypothesis is falsifiable by the proposed experimental design. *Resource Constraints* (weight 15%) evaluates whether the required computation is executable within realistic time and hardware budgets. Assessment is grounded in external, verifiable resources rather than purely textual reasoning; the agent queries biomedical literature, public omics repositories, clinical trial registries, and software package ecosystems to confirm the availability of concrete data and tooling support before passing a hypothesis to the coding validation stage.

Table 7. **External tools and databases queried by the Feasibility Agent to ground its assessment of data availability, technical readiness, and clinical context for each candidate hypothesis.** The Feasibility Agent accesses resources across four categories to ensure its verdicts are based on verifiable, real-world evidence. *Literature* tools (PubMed, Semantic Scholar) are queried to identify relevant prior work, verify biological plausibility, and confirm whether suitable analysis methods have been previously published. *Dataset* repositories (GEO, TCGA/GDC) are queried to verify that sufficiently powered cohorts with the required molecular, imaging, and clinical variables exist and are publicly accessible. *Clinical trial* registries (ClinicalTrials.gov) are queried to identify ongoing or completed trials relevant to the hypothesis, informing clinical feasibility and translational readiness. *Technical* resources (PyPI, GitHub, Papers With Code) are queried to confirm the existence, maturity, and maintenance status of required software packages and reference implementations, ensuring that the proposed computational pipeline can be assembled from available, production-ready components.

Figure 5. **Overview of the MCP-based tool orchestration subsystem that mediates between the SAGE coding agent and heterogeneous clinical and computational tools.** The subsystem is implemented as a Model Context Protocol (MCP) client-server architecture providing a unified control plane for tool access across two domain-specific suites. The *Medical Oncology Suite* exposes clinical analysis tools

including survival analysis and treatment selection, operating over structured patient data. The *Histopathology Suite* exposes a registry of validated whole-slide image analysis tools including HoverFast (fast nuclei detection), mitosis detection, tumor segmentation, HoverNet (nuclei segmentation and classification), collagen segmentation, tubule segmentation, TLS detection, SpaTIL (spatial TIL analysis), multiple instance learning (MIL), GigaTIME (large-scale tumor immune microenvironment analysis), foundation models, patch extraction, and image normalization. The *MCP Orchestrator* sits between the coding agent or researcher and the distributed tool servers. Given a natural-language prompt, the Orchestrator analyzes task intent, dynamically selects the appropriate tools from the registered suites, and coordinates their execution over shared patient data and whole-slide images (WSI), returning structured results. Both Claude and GPT model backends are supported as the reasoning core of the Orchestrator. This architecture abstracts tool location, implementation details, and execution logic, enabling scalable, modular, and extensible tool usage for automated hypothesis validation without requiring manual pipeline configuration.

Figure 6. **User interface of the SAGE tool orchestration subsystem responding to a tumor detection request on an input whole-slide image (WSI).** The left panel displays the cohort and slide selector, allowing a user or autonomous agent to choose a patient cohort and select a specific WSI for analysis. The right panel shows the natural-language interaction interface. The user issues the prompt *"Run tumor detection on this whole slide image"*, specifying the target slide and an output directory. The MCP Orchestrator interprets this prompt, identifies the tumor segmentation tool as the appropriate capability, invokes it on the specified WSI, and returns structured results including the input image path, output directory, and a results summary reporting a detected tumor percentage of 1.42% with confirmation that the segmentation model was successfully loaded and executed. The tool output is a binary tumor mask that highlights detected tumor regions while suppressing non-tumor tissue, visible in the slide viewer on the left. This figure demonstrates the user-facing natural-language interface of the orchestration subsystem and its ability to compose prompt interpretation, tool selection, and execution into a single interaction.

Figure 7. **Region-aware patch extraction via the SAGE tool orchestration subsystem, following automated tumor segmentation on a whole-slide image.** Building on the tumor segmentation output shown in Figure 6, the orchestration subsystem composes a spatial reasoning and patch extraction pipeline that overlays the WSI with color-coded patches corresponding to three tissue regions: tumor regions (red), peritumoral regions defined as tissue immediately adjacent to the tumor boundary (green), and remaining non-tumor tissue (yellow). Patch boundaries are overlaid on the original H&E WSI to visualize the spatial distribution of extracted regions across the tissue section. This region-aware decomposition provides structured, spatially referenced inputs for downstream quantitative analyses such as immune cell density estimation, TLS detection, and collagen quantification within and around the tumor compartment. The figure illustrates how the tool orchestration subsystem chains multiple tools (segmentation, spatial reasoning, and patch extraction) in a single prompted workflow without requiring the user or coding agent to manually specify inter-tool data dependencies.

Table 8. **Representative end-to-end example of the SAGE hypothesis generation and validation workflow applied to CXCL13 as a prognostic biomarker in muscle-invasive bladder cancer (MIBC).** The table traces the complete pipeline from user prompt to feasibility assessment for a single SAGE-generated hypothesis, illustrating how each pipeline stage contributes to the final output. The user prompt requested exploration of genetic and clinical prognostic signatures in locally advanced bladder cancer. SAGE generated the hypothesis that high CXCL13 expression improves progression-free interval in MIBC by mediating B cell and T follicular helper (Tfh) cell recruitment and promoting tertiary lymphoid structure (TLS) formation, thereby enhancing antigen presentation and $CD8^+$ T cell activation. The Novelty Critic assigned a score of 4/10, reflecting that while the CXCL13–TLS axis is known, its specific link to spatially organized immune architecture and TLS maturity as distinct dimensions of MIBC prognosis represents a meaningful extension of existing work. Two coding tasks were specified: an immune activation proxy association analysis using TCGA-BLCA (Firehose Legacy) and a spatial immune cell co-localization and TLS maturity analysis using spatial transcriptomics data. Kaplan–Meier and multivariable Cox analyses on TCGA-BLCA demonstrated consistent and statistically significant survival benefits for CXCL13-high patients (HR $\approx$ 0.62 for PFS and HR $\approx$ 0.64 for OS, both $p < 0.01$), supporting the hypothesis with a Feasibility Score of 8/10. The evaluator noted that direct assessment of the immune activity-proxy sensitivity component and external validation remain outstanding requirements for full clinical translatability.

Table 9. **Context allocation per agent in the SAGE graph-based specific-memory pipeline (GP), specifying which information each agent receives and which upstream outputs are deliberately excluded.** In the GP architecture, each agent receives a customized context containing only the information necessary for its specific task, rather than the full conversation history exposed in the chat-based shared-memory pipeline (CP). This design minimizes token consumption, avoids context dilution from redundant information, and maintains each agent's focus on its defined reasoning scope. Information is excluded when it has

already been integrated and synthesized by a predecessor agent, making it redundant for the receiving agent's task. For example, the *Ontologist* excludes the original user query because the generated paths already encode the query's intent in structured form. The *Scientist* excludes the raw paths because the Ontologist's output already refines and interprets them into biologically grounded terms. The *Novelty Critic* and *Coding Instructions Agent* receive only the final expanded hypothesis, as all earlier conceptual outputs have been synthesized into it by the Hypothesis Expansion Agent. The *Summary Agent* receives only the validation results, final hypothesis, and novelty score, which together contain all information needed for structured reporting. This selective context allocation reduces mean prompt token usage by 57.5% relative to the CP pipeline while maintaining comparable hypothesis novelty (see Table 1).

Table 10. **Composition of the ground-truth evaluation dataset and temporal evaluation protocol used to assess the SAGE multi-critic novelty calibration system on 150 scientific proposals spanning five novelty tiers.** The dataset was constructed to cover the full spectrum from transformative scientific advances to demonstrably invalid claims, with 30 proposals per tier. *Category A (Breakthrough)*: Nobel-winning discoveries and paradigm-shifting methods including AlphaFold2, CRISPR-Cas9, Transformer, ResNet, and GAN; ground-truth novelty scores 8.5–10.0. *Category B (High Novelty)*: significant advances that substantially extend prior work including GAT, BERT, ViT, and SAM; scores 7.0–8.5. *Category C (Moderate)*: meaningful but non-paradigmatic contributions such as novel combinations of existing techniques; scores 5.0–7.0. *Category D (Incremental)*: standard applications and routine extensions such as YOLO variants and conventional SVM pipelines; scores 3.5–5.0. *Category E (Specious)*: fabricated or methodologically invalid claims including STAP Cells, Arsenic Life, and ICEP2; scores 1.0–3.5. To prevent temporal leakage, each proposal was evaluated exclusively against literature available prior to its original publication or retraction date using time-travel backtesting. Categories A and B were evaluated 1–2 years before publication to simulate prospective novelty assessment; Categories C and D were evaluated at the publication year; Category E proposals were evaluated post-retraction to ensure refutation evidence was available.

Deliberation dynamics and literature retrieval behavior of the SAGE multi-critic novelty evaluation system across the five novelty tiers of the ground-truth evaluation dataset ($n = 30$ proposals per tier). Deliberation is triggered when the standard deviation of initial critic scores exceeds SD > 1.0, initiating an iterative Bayesian debate among the Prover, Verifier, and Judge agents. *Debate triggered* reports the fraction of proposals in each tier for which deliberation was activated. *Rounds (avg.)* reports the mean number of debate rounds completed before convergence (SD < 0.5) or the three-round maximum. *Refs. retrieved* reports the mean number of literature sources retrieved per proposal during deliberation (mean $\pm$ SD). *Contradictions found* reports the mean number of retrieved references that directly contradicted or refuted the hypothesis being assessed (mean $\pm$ SD). Deliberation is triggered most frequently for extreme tiers: Category A (76.7%), where critics initially disagree on whether contributions are incremental or genuinely paradigm-shifting, and Category E (83.3%), where critics disagree on whether claims are plausible or fabricated. Breakthrough proposals (A) retrieve few references with minimal contradiction, consistent with genuine gaps in prior literature, whereas specious proposals (E) exhibit the highest contradiction rates (8.7 ± 2.9 per proposal), demonstrating that the system reliably distinguishes the absence of supporting evidence from the presence of refuting evidence. Incremental proposals (D) retrieve the most references overall (9.4 ± 3.1), reflecting their strong overlap with existing indexed work.